

LIU HENGZHI

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EDUCATION

National University of Singapore <i>Bachelor of Computing in Computer Science (Tuition Grant)</i>	2026 – June 2029
Temasek Polytechnic, Singapore <i>Diploma in Biomedical Engineering with Merit</i> Awarded Polytechnic Scholarship, Course Bronze Medalist, and Director's List honors.	2023 – 2026

SKILLS

Programming Languages: Python, C++, C, SQL

Systems & Software: Embedded Systems, Real-time Systems (FreeRTOS), LLM Agents, System Design Basics, Firmware Development

Tools & Frameworks: Git/GitHub, Linux (basic), FastAPI, React, MySQL, CAD (Fusion 360, PTC Creo), Adobe Suite

TECHNICAL PROJECTS

DTx Cognitive SaaS (ADHD Therapeutics) <i>Full-Stack Developer</i> - Built a B2B clinical assessment platform that automatically adjusts test difficulty based on response severity. - Integrated React front-end interfaces with FastAPI backend services to establish clean real-time data flows. - Deployed an AI assistant via Gemini API to analyze telemetry and generate reports.	2026 <i>React, FastAPI, AWS Aurora, LLM Agents</i>
Smart Mahjong Play-Board <i>Independent Systems Developer</i> - Built a smart play-board prototype by designing custom PCBs and integrating microcontrollers. - Implemented game interaction logic using C++ to capture real-time board states.	2026 <i>C++, ESP32, PCB Design, Embedded Firmware</i>

WORK EXPERIENCE

National University of Singapore (iHealthtech) <i>Research Assistant Intern</i> - Ensured high test consistency by strictly executing standardized experimental procedures. - Maintained routine laboratory equipment calibration to secure precise physical measurement data. - Guaranteed research repeatability by establishing structured records for all testing workflows.	Mar 2025 – Jul 2025 <i>Singapore</i>
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CO-CURRICULAR ACTIVITIES

WorldSkills Singapore (Additive Manufacturing) Won 1st Place (TP Selection Round) by modeling and optimizing 3D printable products.	2024
Electronic Design & Robotics Competitions Developed embedded logic systems and real-time control algorithms for robotics challenges.	2024